

Scope

The analysis plan working draft is meant to outline all of the analysis that will be included in the final report. Note that the final report will also include documentation related to data provenance, which is not outlined here.

Format

Each section of the analysis plan should include the following, **as specifically as possible**:

- What questions will it attempt to answer?
- What data elements will be needed?
- What data sources will they be derived from and exactly what variables will be used?
- What visualizations will be produced?
- What tables will be produced?
- What statistical tests will be performed? (what assumptions will be tested in order to justify the use of the test?)
- What types of conclusions do we expect to draw?

Note on consistency: To whatever extent is reasonable, we should be consistent in our choice of statistical tests and visualization methods across various sections, **when the same conditions apply**, rather than each picking our favorite.

Collaboration approach

Each section has an identified point person who will take the lead on drafting content. All team members are encouraged to leave comments, questions, and suggestions as the draft evolves, using the comment function. Topics that get too complex for this method should be discussed directly via Slack or in a meeting.

Analysis plan

Section 1. Demographic comparison of target and benchmark regions

This section builds upon the analysis that was performed during benchmark region selection. It compares characteristics such as population size, industry makeup, income

statistics, and other characteristics that don't relate directly to charitable donations or non-profit organizations.

Basic Question: How do the demographic and economic characteristics of the Black Hills region compare with Benchmark Regions?

***Ref:** The benchmark region analysis that was developed during the benchmark region selection process.

Analysis Plan Table:

1. How does the population size of the Black Hills region compare with benchmark regions?

Data Elements Needed	Available Data Source	Visualization	Table (optional)	Expected Conclusion
Population totals	*American Community Survey (ACS)	Bar chart comparing population across regions	Table of population by region	Whether the BH population base is comparable to benchmark regions

Expected insight: Whether nonprofit comparisons should be adjusted for population.
Whether nonprofit counts should be normalized.

2. How large are the benchmark regions geographically compared to the Black Hills region?

Data Elements Needed	Available Data Source	Visualization	Expected Conclusion
Geographic area (square miles)	* Derived from benchmark definition	Map visualization or bar chart	Contextualize nonprofit density comparisons

3. How does household income compare across regions?

Data Elements Needed	Available Data Source	Visualization	Table (optional)	Expected Conclusion

Median household income; Mean Earnings	*American Community Survey (ACS)	Bar chart comparisons	Income comparison table	Whether income levels are lower or higher, affecting capacity for charitable giving.
---	----------------------------------	-----------------------	-------------------------	--

4. What is the labour market strength in the Black Hills compared to benchmark regions?

Data Elements Needed	Available Data Source	Visualizations	Table (optional)	Expected Conclusion
Unemployment Rate. Population 16+ in Labor Force Civilian Employed Population	*American Community Survey (ACS)	Clustered bar chart and Bar chart	Labor force statistics table	Assess economic stability affecting nonprofit demand

5. How does occupational structure differ between regions?

Data Elements Needed	Available Data Source	Visualization	Table (optional)	Statistical Test	Expected Conclusion
Occupation distribution (Management/Business/Science; Service; Sales/Office; Natural Resources/Construction)	*American Community Survey (ACS)	Stacked bar chart showing occupational distribution	Occupation distribution table	Chi-square test for occupational distribution differences	Identify differences in economic base that may affect philanthropy

6. How do household structures differ across regions?

Data Elements Needed	Available Data Source	Visualization	Table (optional)	Expected Conclusion
----------------------	-----------------------	---------------	------------------	---------------------

Household counts and earning households (Total Households, Households with Earnings, Households with Retirement Income.)	*American Community Survey (ACS)	Clustered Bar chart for household counts and earning households	Household characteristics table	Determine income-earning capacity and donor base; Identify potential presence of older donor populations
---	----------------------------------	---	---------------------------------	---

7. How does household income distribution differ across regions?

Data Elements Needed	Available Data Source	Visualization	Table (optional)	Statistical Test	Expected Conclusion
Income tiers: Low (<\$50k), Middle (\$50k-\$149k), High (\$150k+)	*American Community Survey (ACS)	Stacked bar chart	Income distribution table	Chi-square test of distribution differences	Determine income structure per region

Section 2. Donor characterization

This section describes tax-reported donor behavior using IRS SOI data, focusing on how charitable giving varies across regions at conditional on income levels.

(Note: IRS Statistics of Income (SOI) county-level individual income tax data are currently available only through tax year 2022. This dataset captures only one year (2022) within the intended study period (2022–2024))

1. How do donor participation rates compare across regions within income groups?

Variables	Dataset	Analysis Method	Presentation
- Total Returns - Returns with Charitable Contributions (Count) - Total Adjusted Gross Income (AGI) Group	IRS SOI CBSA/County Data (Tax Year 2022)	- Participation rate = Returns with Charitable Contributions (Count)/Total Returns - Stratify by AGI group	- Boxplot of participation rate by region (within income group) - Table of mean and median

County FIPS code		- Compare participation rate across regions - Descriptive comparison (mean and median participation rate per region and AGI group)	participation rate by region
--	--	---	------------------------------

Expected Conclusions

- How donor participation rates in Black Hills compare descriptively to benchmark regions within the same income group, and whether any observed differences are consistent across income groups or concentrated in specific AGI Groups.

2. How does the average charitable contribution per donor compare across regions within income groups?

Variables	Dataset	Analysis Method	Presentation
- Total Charitable Contributions - Returns with Charitable Contributions - AGI Group - County FIPS code	IRS SOI CBSA/County Data (Tax Year 2022)	- Average contribution per donor = Total Charitable Contributions>Returns with Charitable Contributions - Stratify by AGI group - Log transform if distribution is skewed - Descriptive comparison (mean and median average gift per region and AGI group)	- Boxplot of average contribution by region - Line plot: income group vs average contribution by region - Summary statistics table

Expected Conclusions

- How average gift size in Black Hills compares descriptively to benchmark regions within the same income group, after accounting for income level.

- Identifies whether any observed gap is consistent across income groups or concentrated at a particular group.

3. How do charitable contributions relative to income compare across regions?

Variables	Dataset	Analysis Method	Presentation
• Total Charitable Contributions • Total Adjusted Gross Income (AGI) • AGI Group • County FIPS code	• IRS SOI CBSA/County Data (Tax Year 2022)	• Giving ratio = Total Charitable Contributions/Total Adjusted Gross Income (AGI) • Stratify by AGI group • ZIP → County FIPS • Compare distributions across regions • Descriptive comparison (mean and median giving ratio per region and AGI group)	• Boxplot of giving ratio by region • Heatmap of region × income group

Expected Conclusions

- How the giving ratio in Black Hills compares descriptively to benchmark regions within the same income group.

4. Are regional differences in total giving driven by participation or contribution size?

Variables	Dataset	Analysis Method	Presentation
• Total Returns • Returns with Charitable Contributions • Total Charitable Contributions • AGI Group • County FIPS code	• IRS SOI CBSA/County Data (Tax Year 2022)	• Participation rate = Returns with Charitable Contributions (Count)/Total Returns • Average contribution per donor = Total Charitable Contributions/Returns with Charitable Contributions	• Decomposition bar chart (participation vs average contribution) • Side-by-side comparison plots

		• Compare contribution drivers across regions • Assess whether differences arise from participation vs contribution size	
--	--	---	--

Expected Conclusions

- Whether the Black Hills giving profile is driven primarily by a broader donor base (participation) or by higher per-donor amounts (intensity).

5. Does the relationship between income and charitable giving differ across regions?

Variables	Dataset	Analysis Method	Presentation
• Total Charitable Contributions • Returns with Charitable Contributions • Total AGI • AGI Group	• IRS SOI CBSA/County Data (Tax Year 2022)	• Participation rate = Returns with Charitable Contributions (Count)/Total Returns • Average contribution per donor = Total Charitable Contributions/Returns with Charitable Contributions • Giving ratio = Total Charitable Contributions/Total Adjusted Gross Income (AGI) • Evaluate trends across income groups • Compare slopes across regions at region level	• Line plot: income group vs giving metrics (by region) • Comparative trend visualization

Expected Conclusions

- How the income-giving gradient in Black Hills compares visually to benchmark regions, and whether higher-income donors appear to drive giving disproportionately in certain regions.

6. Is rurality associated with donor participation rates across counties?

Variables	Dataset	Analysis Method	Presentation
- Donor participation rate by AGI group - Rural-Urban Continuum Code (RUCC) - County FIPS code	- IRS SOI CBSA/County Data (Tax Year 2022) - USDA Rural-Urban Continuum Codes	- Spearman rank correlation between RUCC code and participation rate, pooled across all counties in the study regions - Simple linear regression (RUCC as predictor, participation rate as outcome), one model per AGI group - Analysis unit: individual counties	- Scatter plot: RUCC code vs. participation rate (by AGI group) - Correlation coefficient table

Expected Conclusions

- Whether counties with higher rurality (higher RUCC codes) show systematically lower donor participation rates, and whether this relationship holds consistently across AGI groups.

7. Is nonprofit density associated with donor participation rates across counties?

Variables	Dataset	Analysis Method	Presentation
- Donor participation rate by AGI group - Number of nonprofit organizations per county - Population per county	- IRS SOI CBSA/County Data (Tax Year 2022) - IRS EO-BMF - ACS	- Spearman rank correlation between nonprofit density and participation rate - Simple linear regression (nonprofit	- Scatter plot: nonprofit density vs. participation rate - Correlation coefficient table

County FIPS code		density as predictor), one model per AGI group - Nonprofit density = number of nonprofits / population × 10,000 - Analysis unit: individual counties	
--	--	---	--

Expected Conclusions

- Whether counties with higher nonprofit density show higher donor participation rates, suggesting that supply-side infrastructure (availability of recipient organizations) is associated with giving behavior.

Notes

- SOI data reflect itemized tax returns only
- No statistical significance testing (Q1–Q3, Q5): Billings, Missoula, and Flagstaff each consist of a single county, making distributional tests. Conclusions describe observed patterns and magnitudes, not statistical significance.
- Results represent tax-deductible giving behavior, not total population giving.

Section 3. Non-profit organization characterization

This section describes non-profit organizations including size, budget, funding sources and mission types, with comparison between the target and benchmark regions.

1. Is there a significant difference in the number and density of nonprofit organizations between Black Hills and benchmark regions?

Variables	Datasets	Analysis Method	Presentation
Total number of nonprofit organizations for each	- IRS EO-BMF - American Community Survey	Density = number of nonprofits/	- ANOVA result - Bar graphs of number and

region, identify by EIN Population size		10K population ANOVA	density of nonprofits density by region
---	--	---	--

2. Is there a rural and/or indigenous service gaps within the Black Hills community (compare counties)?

Variables	Datasets	Analysis Method	Presentation
- County (address) - Total number of organizations for each county - Rural-Urban Continuum Codes - Indigenous population	- IRS EO-BMF - American Community Survey - Economic Research Service		Map of nonprofit density per county

3. Is there a significant difference in total revenue between Black Hills and benchmark regions?

Variables	Datasets	Analysis Method	Presentation
- Total Revenue - NTEE filed classification code	IRS EO-BMF	- Normalize revenue by number of nonprofits for each region - ANOVA - With and without hospitals, universities, and political orgs	- Anova result - Bar graph of normalized total revenue for each region

4. Is there a significant difference in total assets between Black Hills and benchmark regions?

Variables	Datasets	Analysis Method	Presentation
- Total Assets - NTEE filed classification code	IRS EO-BMF	- Normalize assets by number of nonprofits for each region - ANOVA - With and without hospitals, universities, and political orgs	- Anova result – any conclusions about statistical significance - Bar graph of normalized total assets for each region

5. Is there a significant difference in nonprofits' dependency on grants vs. other revenue sources between Black Hills and benchmark regions?

Variables	Datasets	Analysis Method	Presentation
- Grants (total amount) - Total revenue - NTEE field classification code	- USAspending .gov - Form 990 - IRS EO-BMF	- Percentage of grants in total revenue - ANOVA - With and without hospitals, universities, and political orgs	- ANOVA result - Bar graph of percentage of grants in total revenue for each region

6. Are there any nonprofit fields under/over-represented in each region?

Variables	Datasets	Analysis Method	Presentation
NTEE field classification code (only use the broad)	IRS EO-BMF	Chi-squared test	- Chi-squared test result - Heatmap of region vs. NTEE fields

category, first letter) • Total revenue • Total asset			• Total revenue and asset by field (bar, pie)
---	--	--	---

7. What are the best predictors of total revenue for Black Hills and each benchmark region?

Variables	Datasets	Analysis Method	Presentation
• Total revenue • % of armed forces • Average annual employment • Average annual wages • % of broad NTEE field classification code	• IRS EO-BMF • American Community Survey • Bureau of Labor Statistics	• Normalize revenue by the number of nonprofits for each region (response variable) • Multiple regression	• Multiple regression result in a table comparing the significance of predictors for each region

8. How has the number of nonprofit organizations and performance changed over time (2022-2024)?

Variables	Datasets	Analysis Method	Presentation
• Number of nonprofit orgs for each region for each year • Population	• IRS EO-BMF	• Multiple regression for each variable controlling for population	• Regression result in a table • Line graph normalized for change in population size

For limited number of organizations (Form 990, EZ, PF):

1. Is there a difference in the revenue sources between Black Hills and benchmark regions?

Variables	Datasets	Analysis Method	Presentation
• Total revenue • Program service revenue • Total contributions • Other contributions (foundation grants etc)	• NCCS • Total contributions for 990, EZ, and PF • Program service revenue from 990 and EZ	• ANOVA	• Stacked bar graph • Pie chart for Black Hills region (communication only)

2. Is there a difference in financial performance of nonprofit organizations between Black Hills and benchmark regions (2022-2024)?

Variables	Datasets	Analysis Method	Presentation
• Total Revenue • Total Expense • Net Asset	• Form 990	• Months of reserves ($\text{Net Assets} \div \text{Total Expenses} \times 12$) • Net margin = $(\text{surplus} = \text{revenue} - \text{expenses}) / \text{total revenue}$ • Multiple regression controlling for population	• Regression result in a table • Line graph

Notes:

1. Form 990 vs. IRS EO-BMF
 - a. IRS EO-BMF: includes all IRS-recognized organizations that are tax-exempt, including 501c3 organizations, private foundations, and political organizations
 - i. Organizations that are not required to file Form 990 are included (e.g. churches)

- ii. Only total revenue, income, and assets are available
- b. Form 990:
 - i. Form 990: filed by organizations with gross receipts over \$200K or total assets over \$500K
 - 1. Shows detailed revenue source and expense data for each organization
 - ii. Form 990-EZ: filed by organizations with gross receipts \$50K ~ \$200K
 - iii. Form 990-PF: filed by private foundations regardless of financial status
 - iv. Form 990-N: filed by organizations with less than \$50K gross receipts
 - 1. No financial details are provided
- 2. NTEE codes: <https://urbaninstitute.github.io/nccs-legacy/ntee/ntee.html>
 - a. Hospitals
 - i. E20 – Hospitals
 - ii. E21 – Community Health Systems
 - iii. E22 – General Hospitals
 - iv. E24 – Specialty Hospitals
 - b. Universities
 - i. B40 – Higher Education
 - ii. B41 – Two-Year Colleges
 - iii. B42- Undergraduate Colleges
 - iv. B43 – Universities
 - v. B50 – Graduate and Professional Schools
- 3. Political organizations
 - a. 501c4 - social welfare, but also often political according to Eric
 - b. 527 – political committees